

National University of Computer and Emerging Sciences



Laboratory Manual #1
for
Programming Fundamentals
(CL 1002)

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Lab Instructor(s)	Ms. Hira Butt Ms. Arfa Masood
Section	B
Semester	Fall 2022

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FAST-NU, Lahore, Pakistan

Introduction

Objectives

After performing this lab, students shall be able to:

- Use university's intranet for different tasks
- Make assignment submissions
- Use flex, xeon and software services.
- Open internet explorer and compose an email
- Use google groups
- Practice with a typing tutor

Problems / Assignments

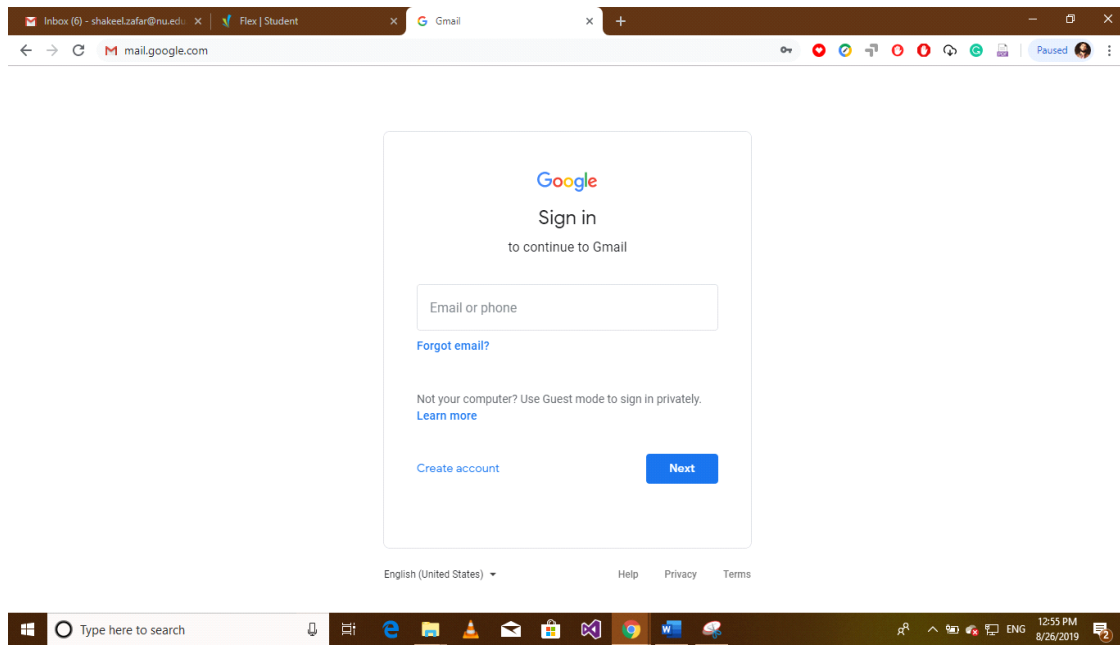
Problem 1: Introduction to various parts of a computer	
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Problem 2: Introduction to Windows environment and some important applications	
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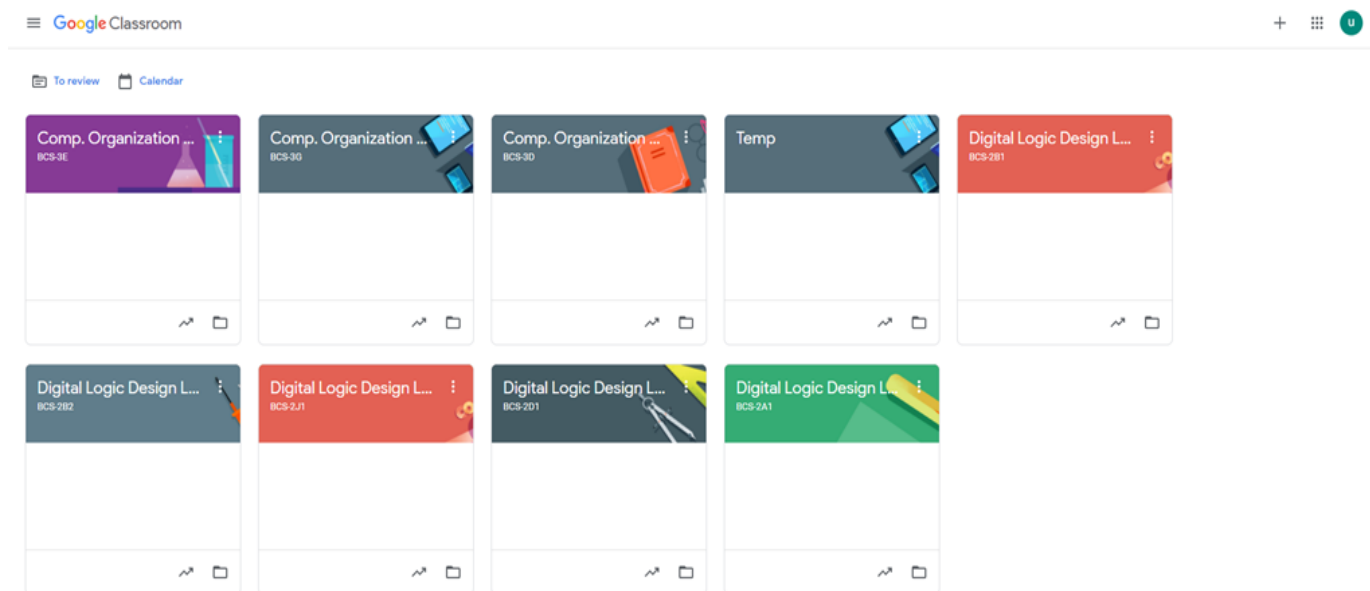
Learn about the concept of a desktop and how to open various applications.

Problem 3: Log on to NU-Mail and check your emails	
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Visit URL: <http://mail.lhr.nu.edu.pk/> OR <https://mail.google.com>



Problem 4: Explore Google Classroom





PF-Lab
1J

Stream

Classwork

People

Grades



Meet

Generate link

Class code

q2bwb3h

Upcoming



Announce something to your class



This is where you can talk to your class

Use the stream to share announcements, post assignments, and respond to student questions



Stream settings



g-22

Stream

Classwork

People

Grades

+ Create



Google Calendar



Class Drive folder



Lab-13

Posted Jun 3



Lab-12

Due May 27, 1:00 PM



Lab static and dynamic binding

Edited May 20



Lab-10

Due May 13, 1:00 PM



Lab-09

Due Apr 30, 11:59 PM



Lab-08

Due Apr 16, 4:00 AM



Lab-07

Due Apr 8, 10:00 PM



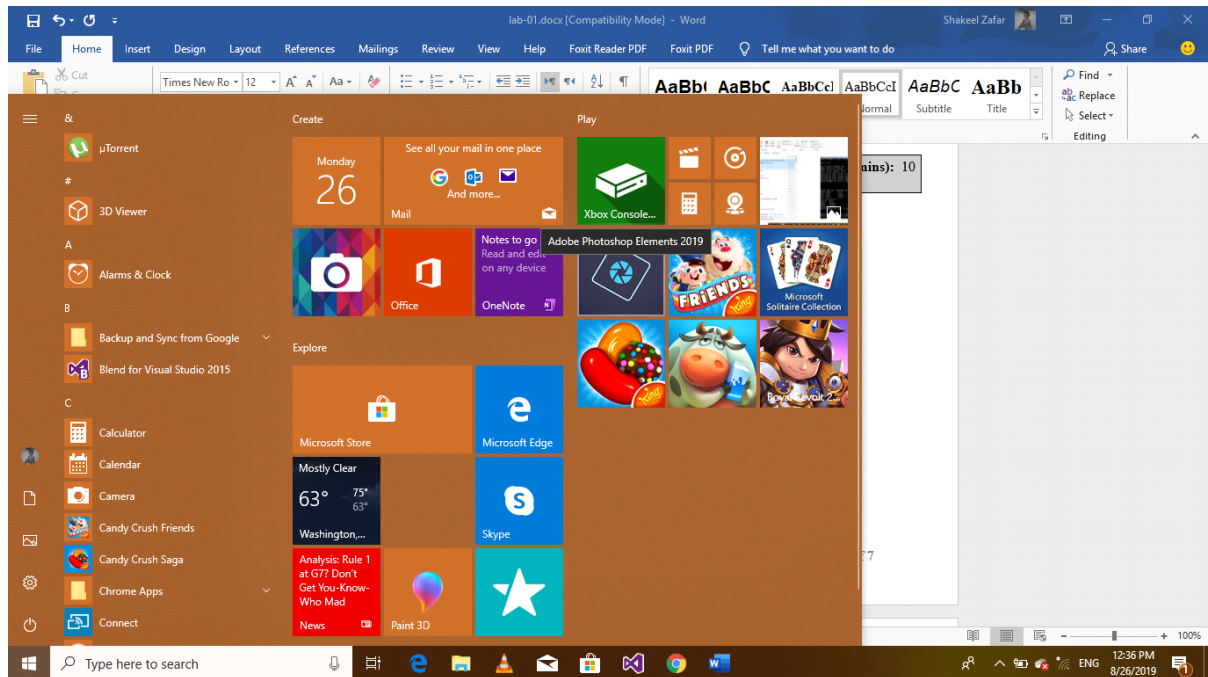
Lab-06 1

Due Apr 5, 9:00 PM

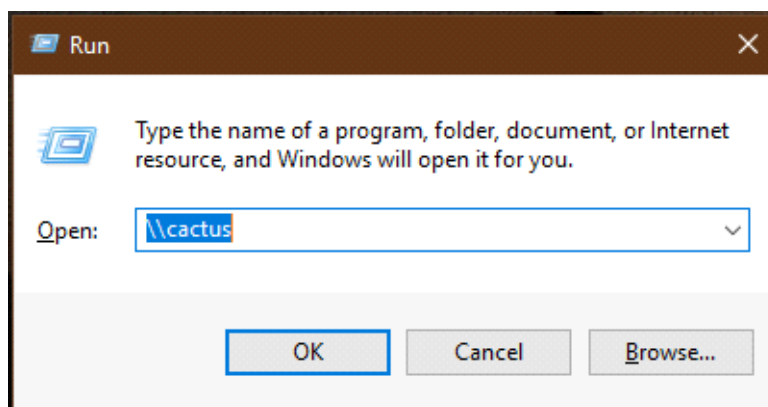
Problem 5: Explore files on xeon server and copy

files there

Steps:



- Go to “Start” and in Search bar enter “Run”
- Click on Run program



- Enter address\\cactus and explore different folders in them.

Windows Security

Enter network credentials

Enter your credentials to connect to: cactus

User name

Password

☐ Remember my credentials

The user name or password is incorrect.

OK Cancel



Windows Security

Enter network credentials

Enter your credentials to connect to: cactus

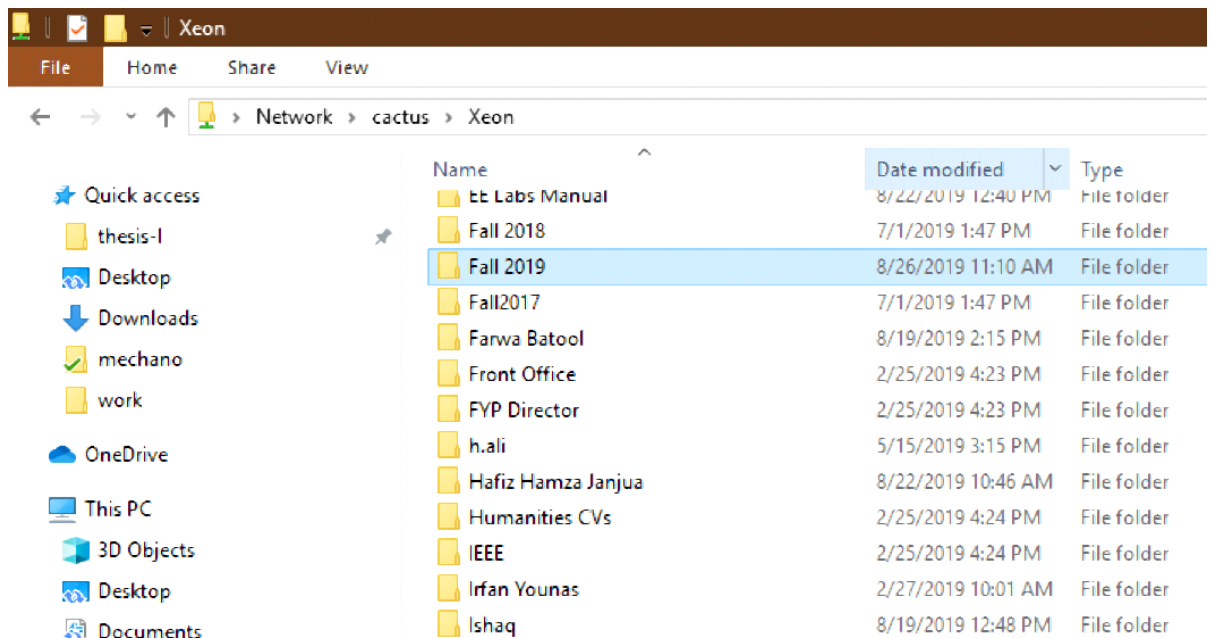
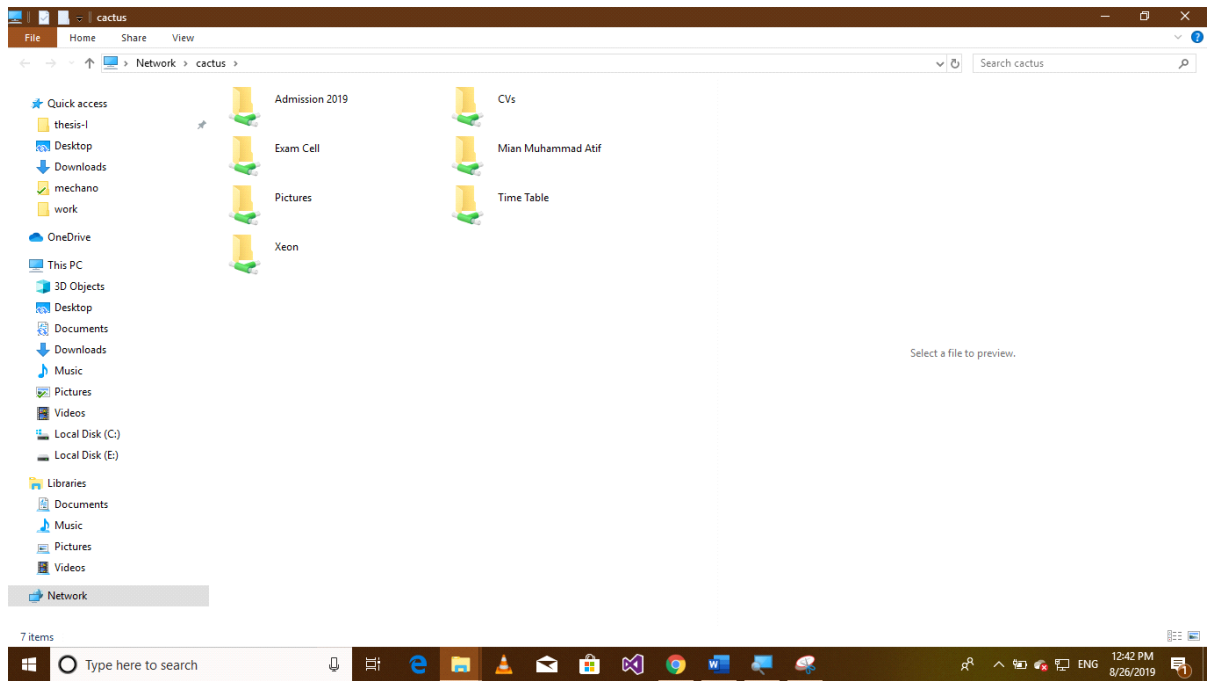
domain\190123

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☐ Remember my credentials

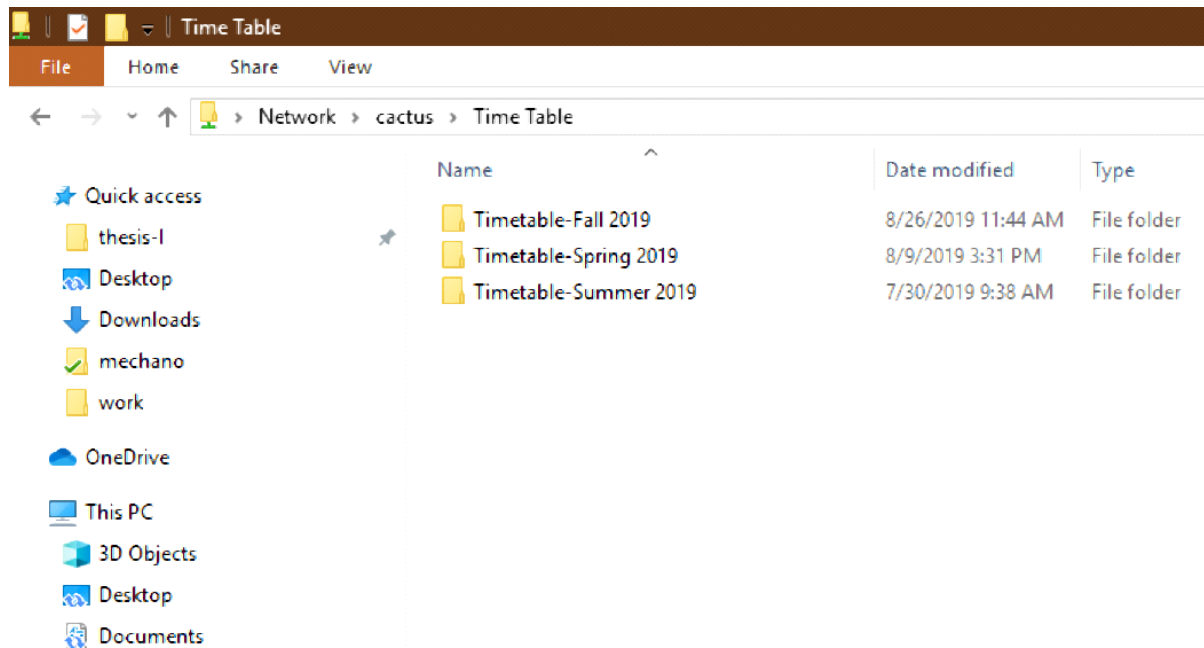
The user name or password is incorrect.

OK Cancel

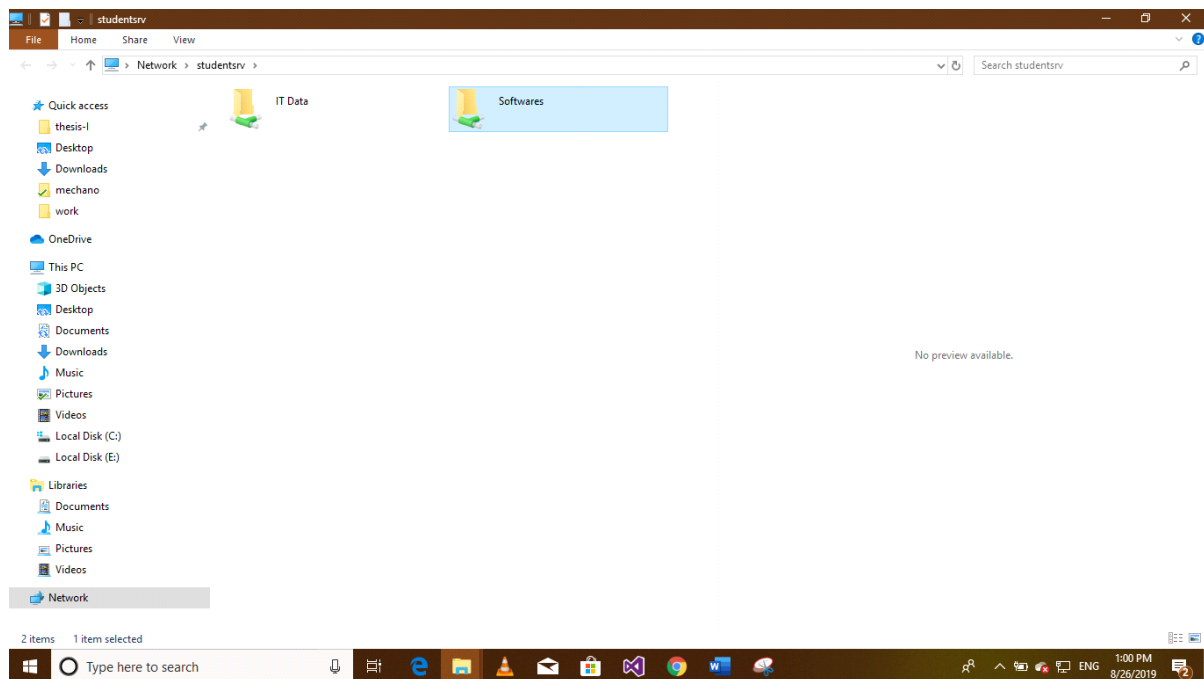


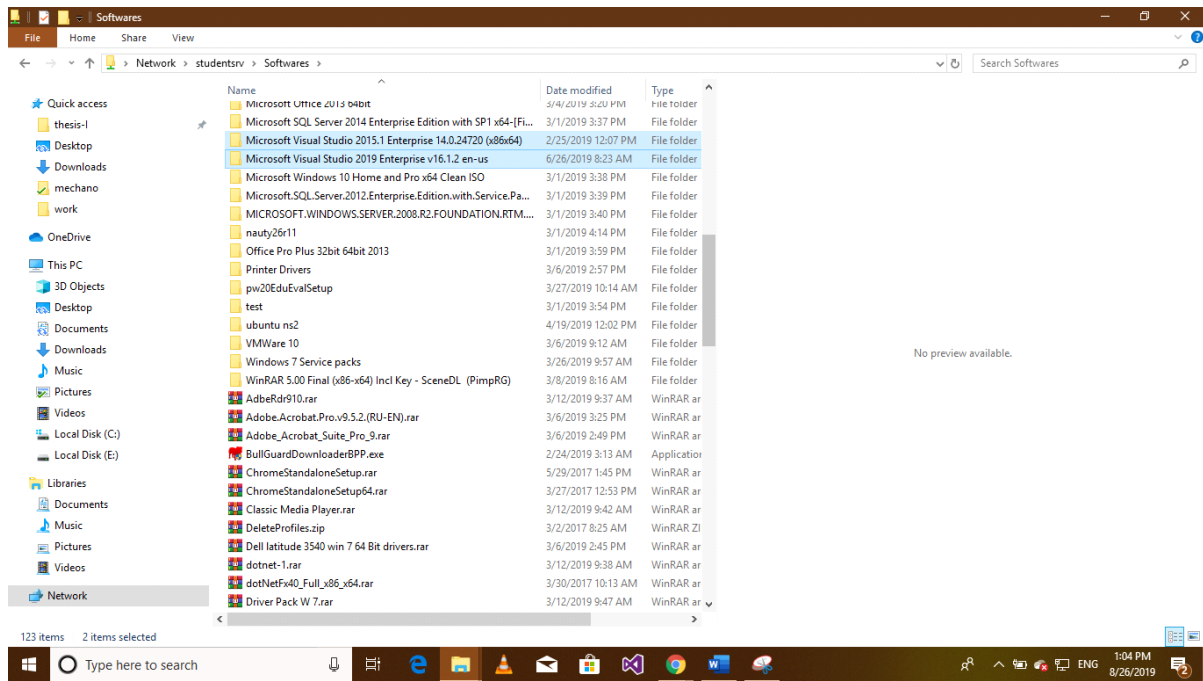
- Learn how to submit the folder.

Problem 5 (a): open Timetable folder to see timetable



Problem 5 (b): open studentsrv folder to have software

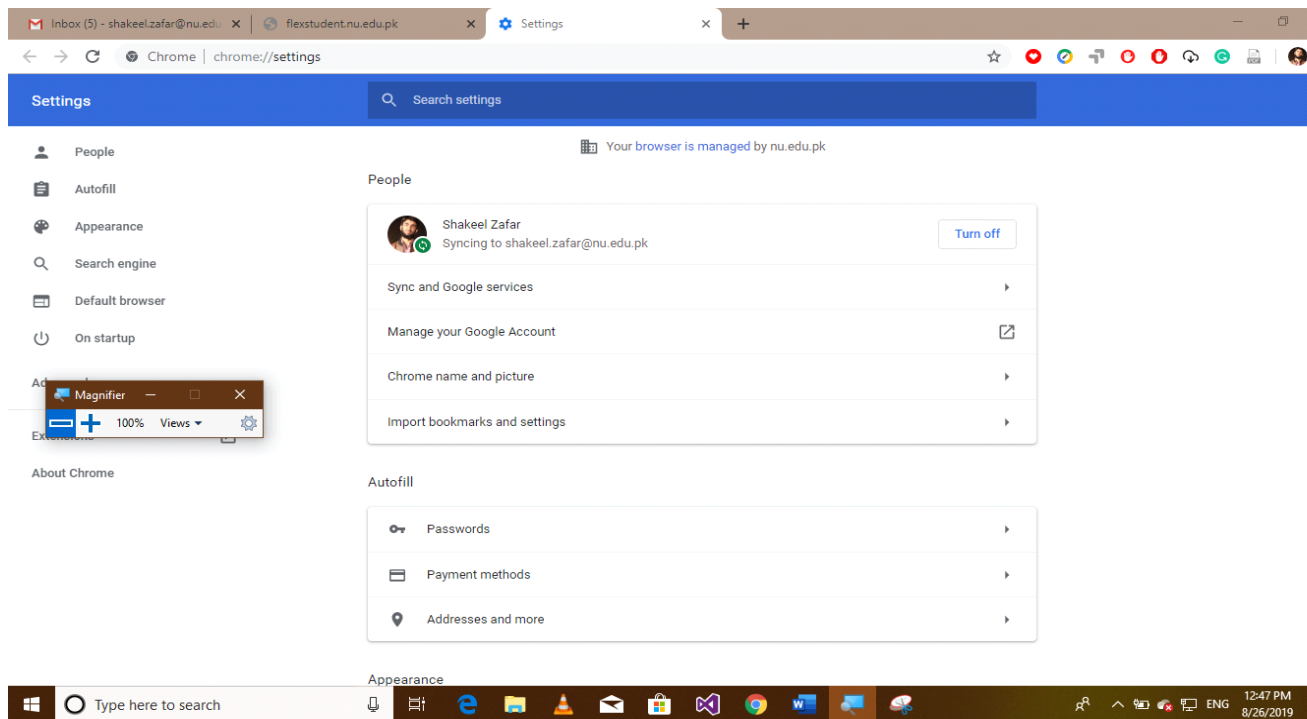




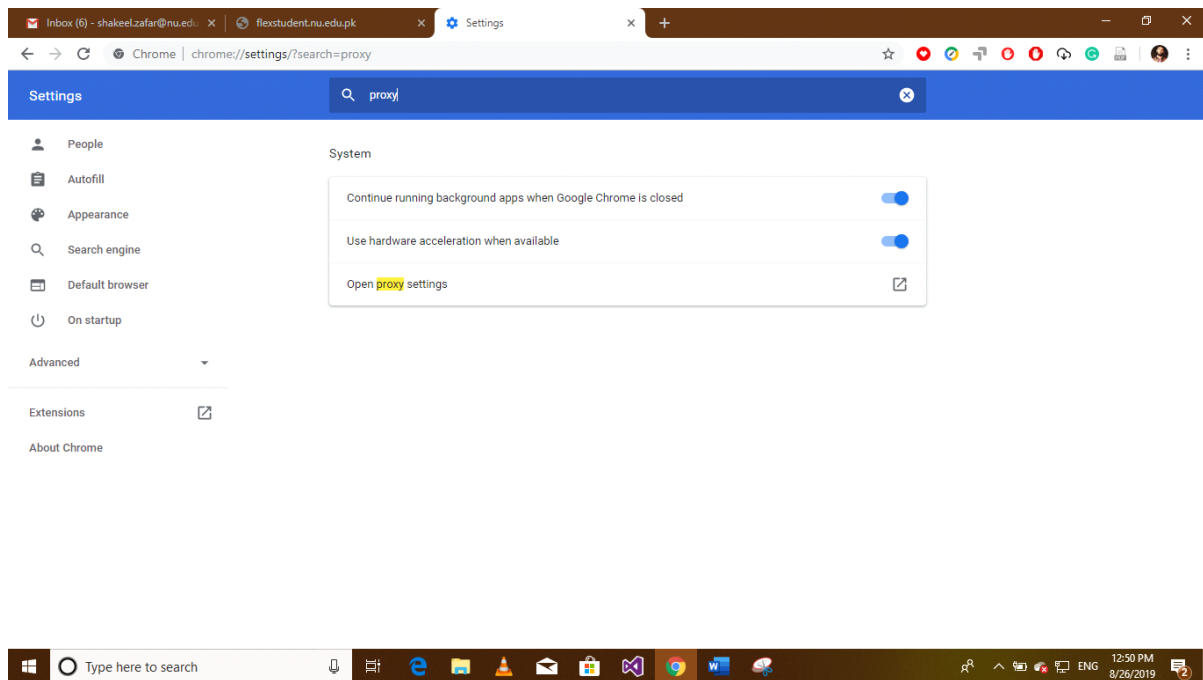
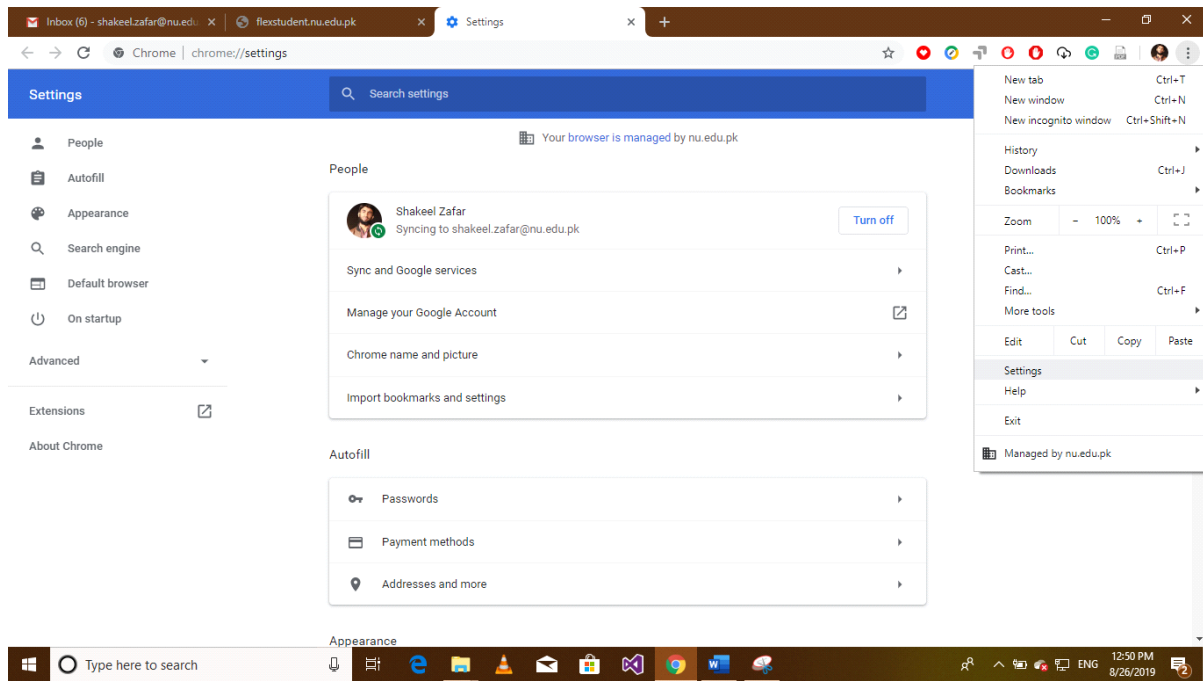
Problem 6: Log onto flex and check your list of courses etc.

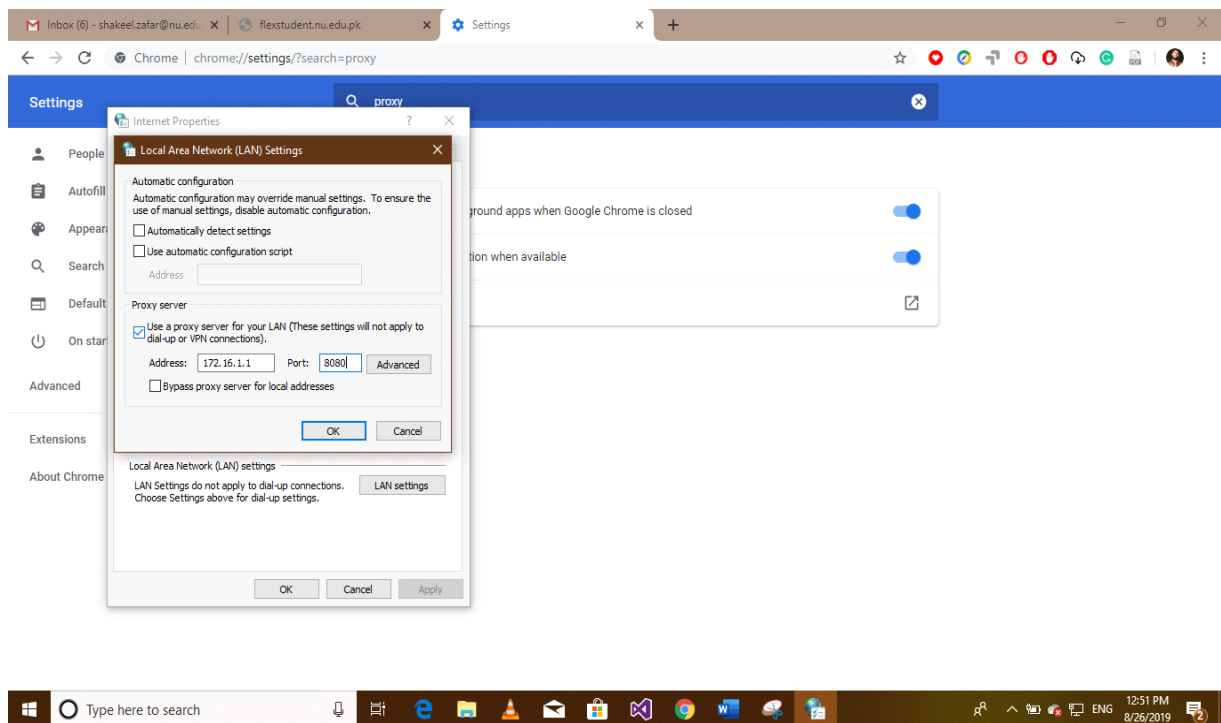
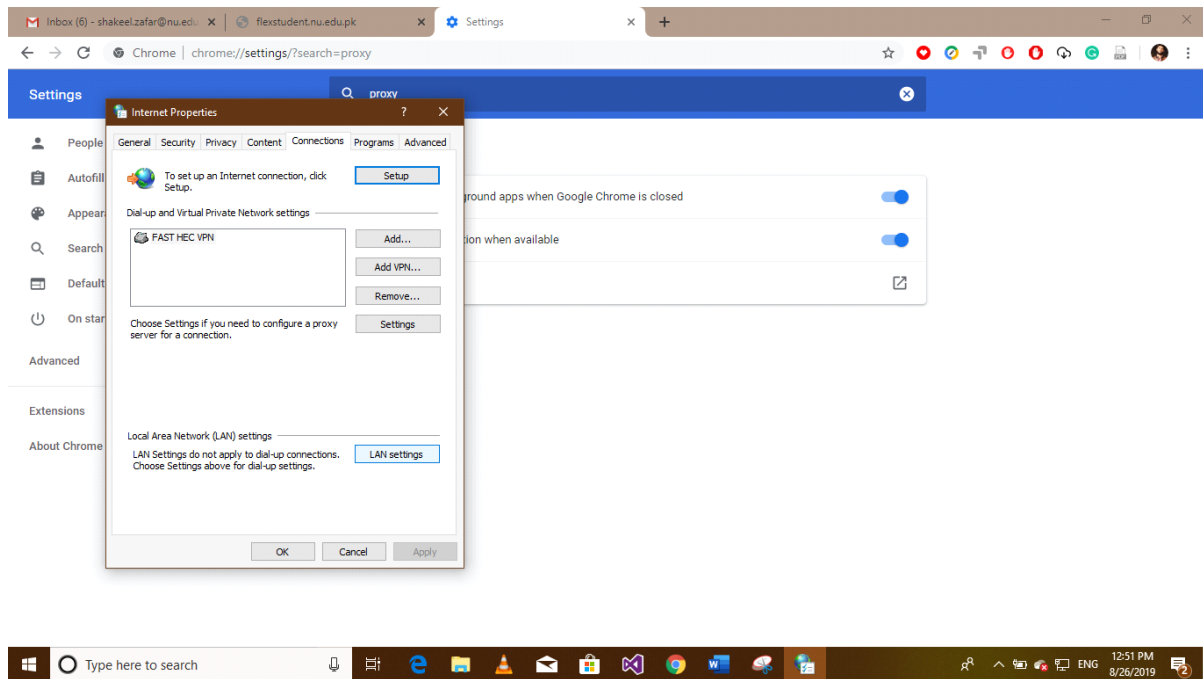
Steps:

-

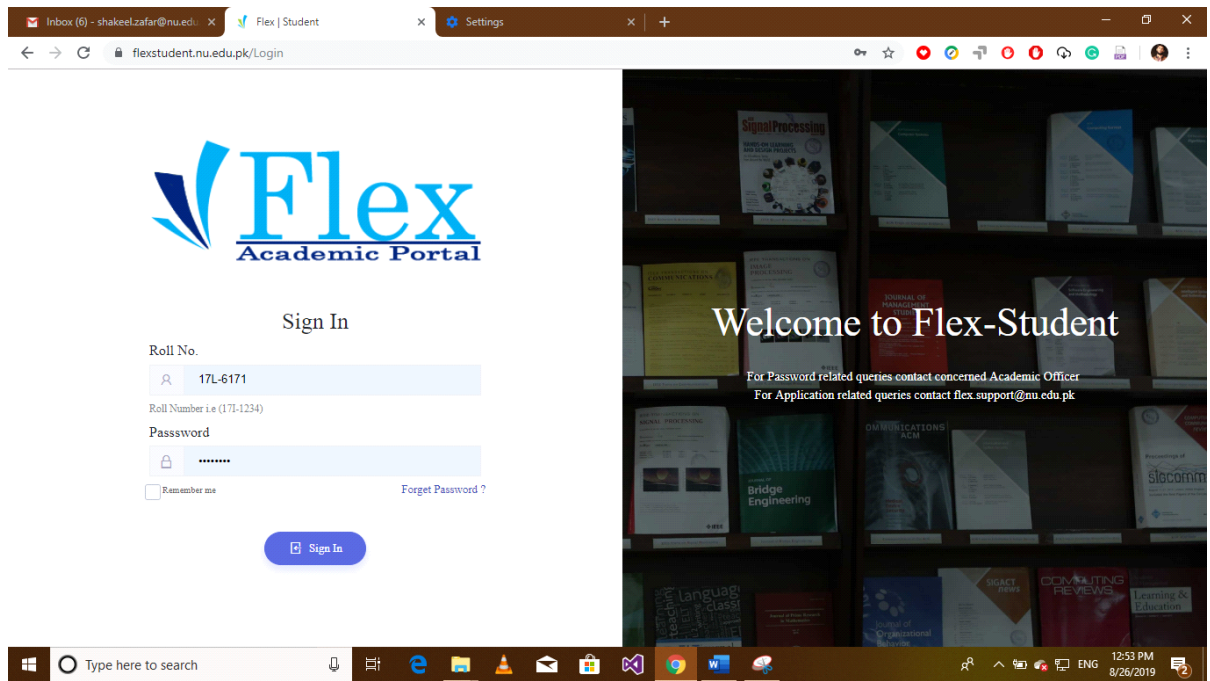


Open google chrome





- Visit url: <http://flexstudent.nu.edu.pk/>



Problem 7: Algorithms

Objectives

The objectives of this part are to cover the following:

- Problem Solving
- What is an algorithm?
- How do we write an algorithm in simple English?

Important Notes

- Be aware that you are asked to write algorithms in simple English language.

Question#1

Write an algorithm that takes as input a list of numbers from the user and then a number to search within it.

Example Input:

List: 12, 17, 3, 44, 77, 2, 1

Number to search: 2

Output: Number 2 exists in the list

Number to search: 21

Output: Number 21 does not exist in the list

Question#2

Write an algorithm that checks whether a list of numbers is sorted in ascending order.

Example Input:

List: 12, 17, 3, 44, 77, 2, 1

Output: List is not sorted in ascending order

List: 12, 17, 44, 77, 91

Output: List is sorted in ascending order.

Question#3

Write an algorithm that finds the max element in a list of numbers.

Question#4

Write an algorithm to measure out exactly **4 liters** of water.

But, of course, you have a problem. You have two containers each **3 and 5 liters** in volume respectively.

Each container has no other marking except for the fact that it only provides its known volume. Using a running tap to fill them how would you measure out exactly **4 liters**?

Question#5

Suppose you have lost your watch. After brainstorming you remembered that last time you placed the watch in the cupboard that has 5 shelves. Now Write a pseudo code to search your watch in each shelf of the cupboard.

Problem 8: Open MS word and experiment with basic commands	Estimated completion time (mins): 20
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- Open MS Word and write a paragraph to introduce yourself. Experiment with the following:
- Cut (ctrl X), paste (ctrl v), copy (ctrl c), save (ctrl s), save as, bold (ctrl b), italic (ctrl i), underline (ctrl u)
- Use the basic formatting commands, such as change font size, paragraph spacing

Problem 9: Practice typing and placement of fingers for typing	Ongoing
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Learn the placement of fingers for typing at:

<http://www.typingcourse-online.com/EN/Aspx/SelectExerciseWithCharacters.aspx>

Problem 10: RoboZZle	Ongoing
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Go to the following website:

<http://www.robozzle.com/>

- Make groups of two and learn to play this game.
- You have to clear the first three levels of this game
 - Stairs
 - Sh
 - Right on red

If you have a smart phone, download the game on your phone and play it. Enjoy!

READING MATERIAL/REFERENCE

Please see the page:

<http://support.microsoft.com/kb/126449>

It has very useful information on keyboard shortcuts to be used